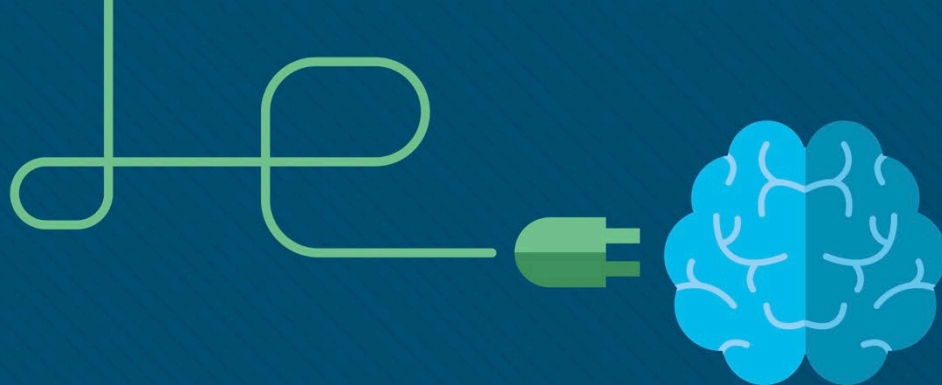




Module 9: FHRP Concepts

Switching, Routing and Wireless
Essentials v7.0 (SRWE)



Module Objectives

Module Title: FHRP Concepts

Module Objective: Explain how FHRPs provide default gateway services in a redundant network.

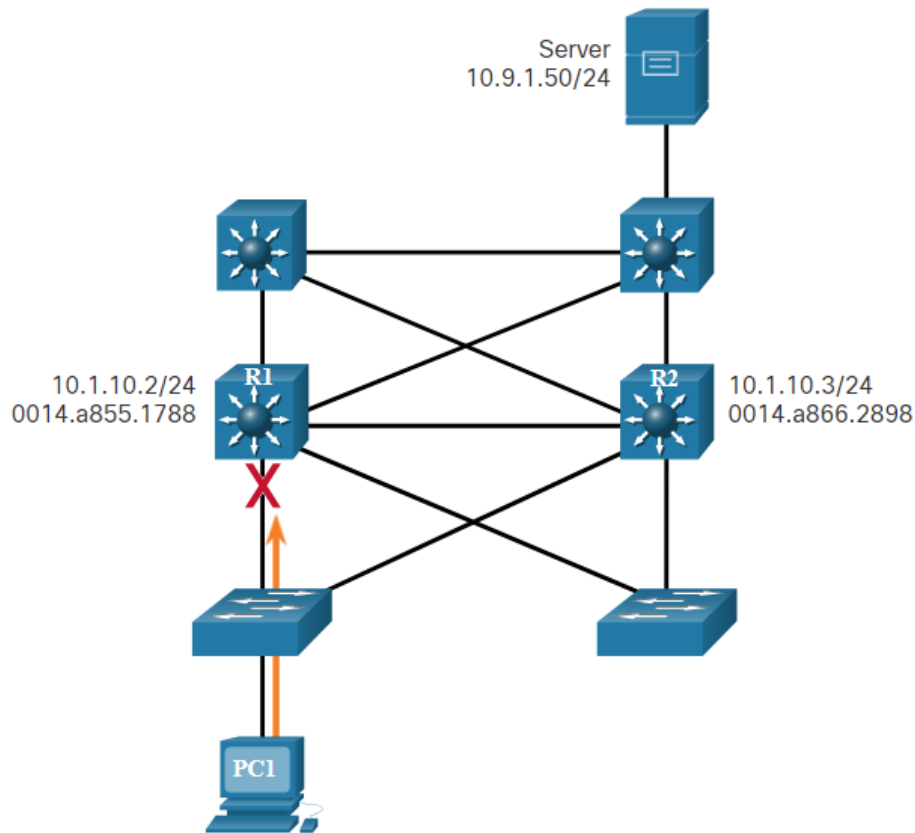
Topic Title	Topic Objective
First Hop Redundancy Protocols	Explain the purpose and operation of first hop redundancy protocols.
HSRP	Explain how HSRP operates.

9.1 First Hop Redundancy Protocols

First Hop Redundancy Protocols

Default Gateway Limitations

- If the default gateway router interface fails, LAN hosts lose outside LAN connectivity.
- This occurs even if a redundant router or Layer 3 switch that could serve as a default gateway exists.
- First hop redundancy protocols (FHRPs) are mechanisms that provide alternate default gateways



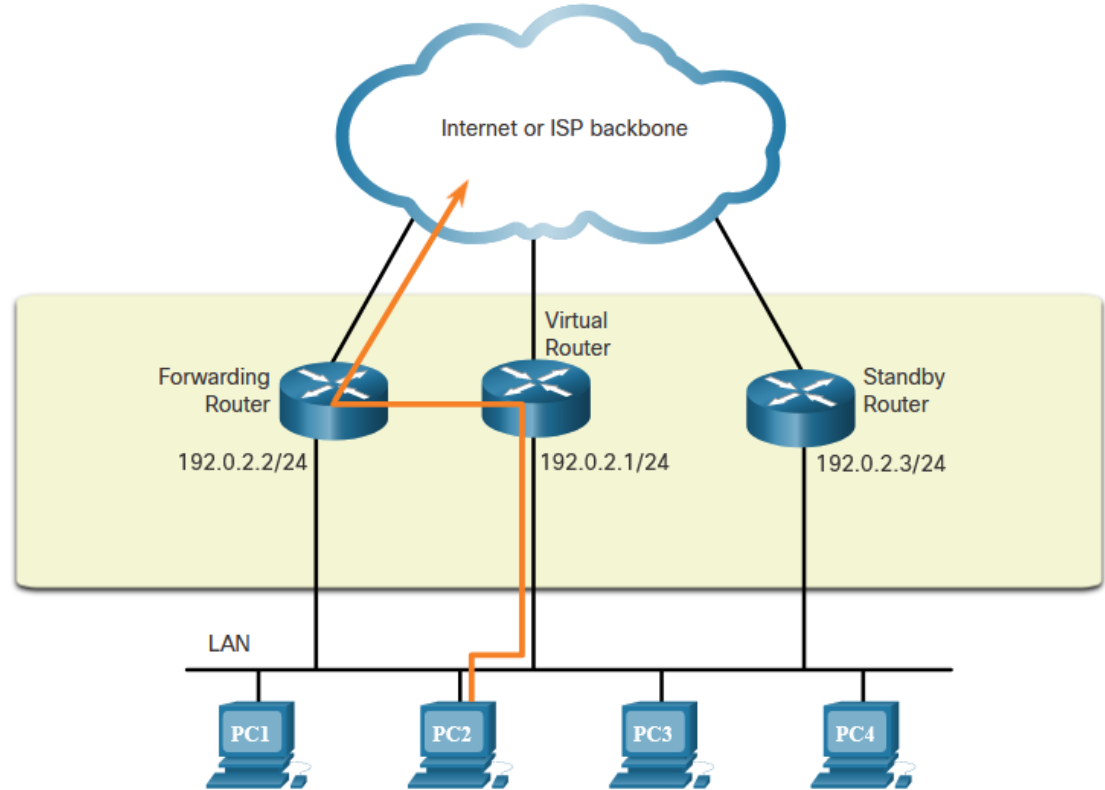
First Hop Redundancy Protocols

Virtual Router

One way to prevent a single point of failure at the default gateway is to implement a **virtual router**

Its IPv4 address is the **Default Gateway**

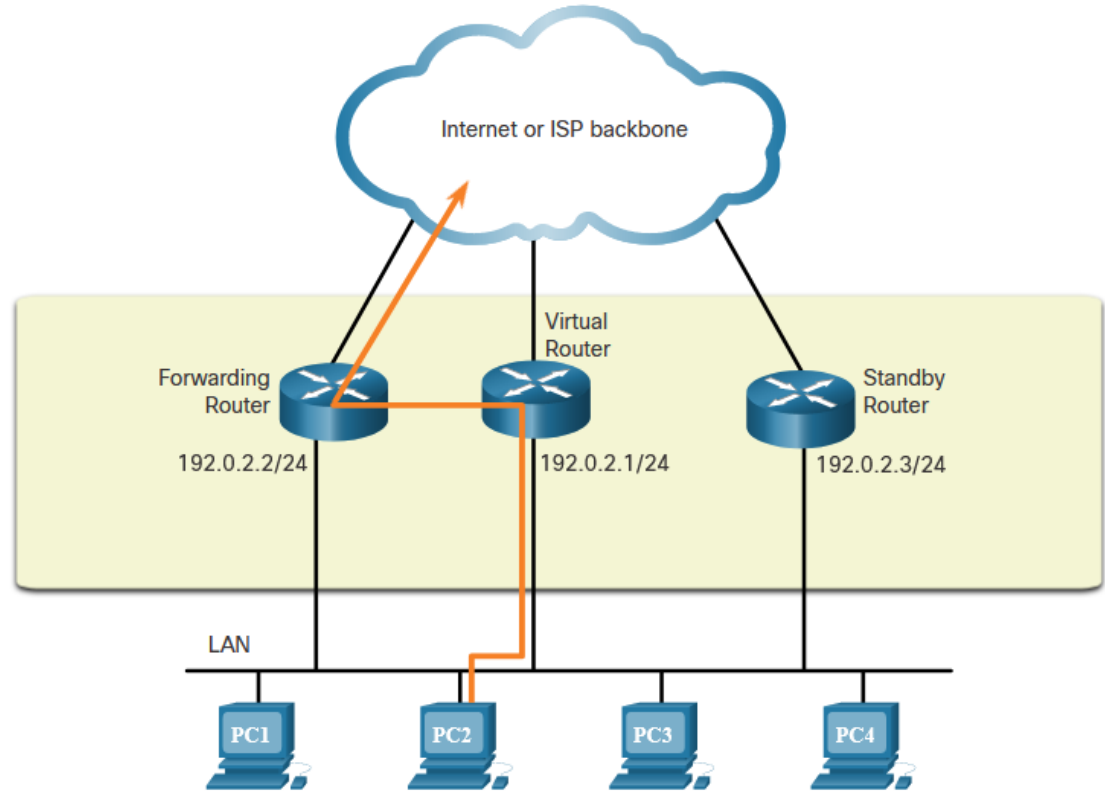
The ARP resolution returns the MAC address of the virtual router



First Hop Redundancy Protocols

Virtual Router

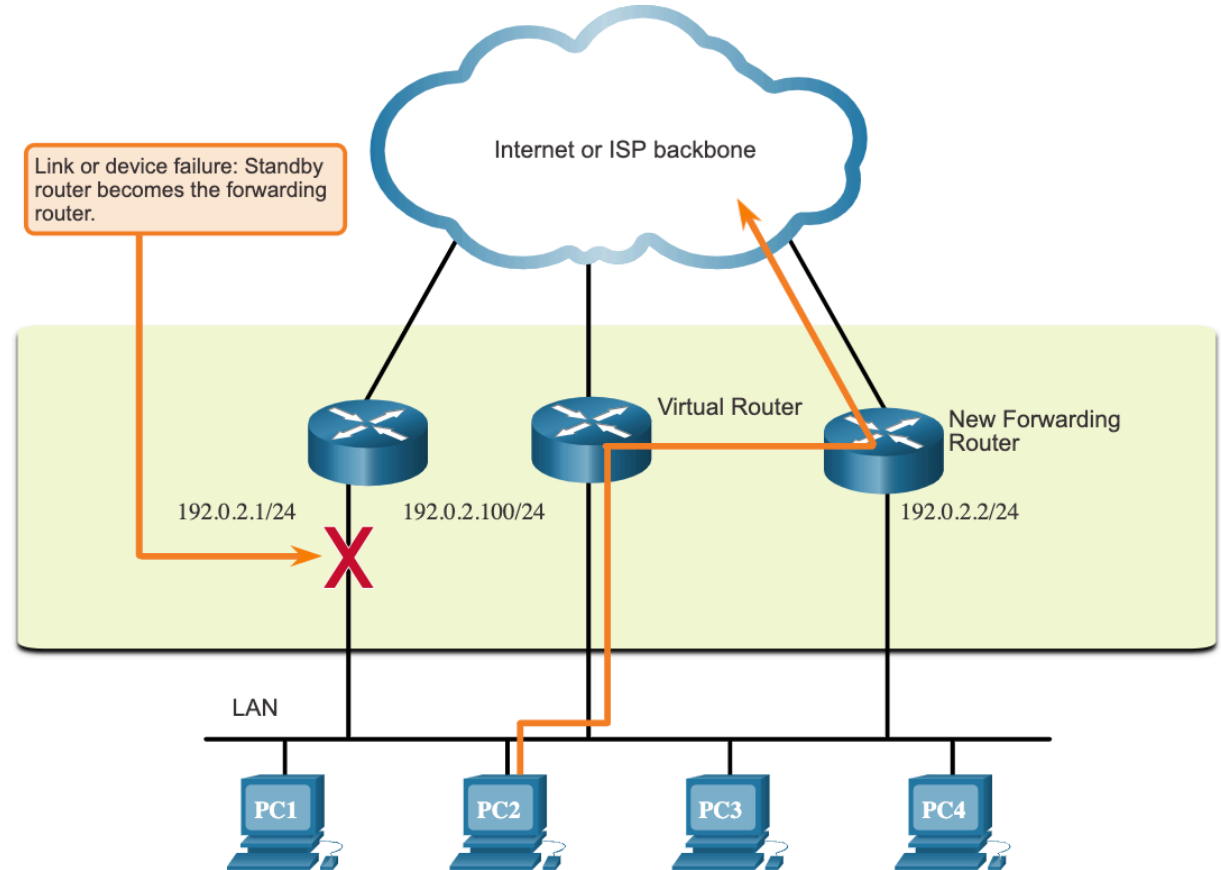
A redundancy protocol decides which physical router is active



First Hop Redundancy Protocols

Steps for Router Failover

- Standby Router
 - Active Router
1. The standby router stops seeing Hello messages from the forwarding router.
 2. The standby router assumes the role of the forwarding router.
 3. Because the new forwarding router assumes both the IPv4 and MAC addresses of the virtual router, the host devices see no disruption in service.



First Hop Redundancy Protocols

FHRP Options

FHRP Options	Description
Hot Standby Router Protocol (HSRP)	HSRP is a Cisco-proprietary FHRP that is designed to allow for transparent failover of a first-hop IPv4 device. HSRP is used in a group of routers for selecting an active device and a standby device. The active device is the device that is used for routing packets; the standby device is the device that takes over when the active device fails, or when pre-set conditions are met.
HSRP for IPv6	This is a Cisco-proprietary FHRP that provides the same functionality of HSRP, but in an IPv6 environment. An HSRP IPv6 group has a virtual MAC address derived from the HSRP group number and a virtual IPv6 link-local address derived from the HSRP virtual MAC address. Periodic router advertisements (RAs) are sent for the HSRP virtual IPv6 link-local address when the HSRP group is active. When the group becomes inactive, these RAs stop after a final RA is sent.
Virtual Router Redundancy Protocol version 2 (VRRPv2)	This is a non-proprietary election protocol that dynamically assigns responsibility for one or more virtual routers to the VRRP routers on an IPv4 LAN. This allows several routers on a multiaccess link to use the same virtual IPv4 address. In a VRRP configuration, one router is elected as the virtual router master, with the other routers acting as backups, in case the virtual router master fails.
VRRPv3	This provides the capability to support IPv4 and IPv6 addresses. VRRPv3 works in multi-vendor environments and is more scalable than VRRPv2.
Gateway Load Balancing Protocol (GLBP)	This is a Cisco-proprietary FHRP that protects data traffic from a failed router or circuit, like HSRP and VRRP, while also allowing load balancing (also called load sharing) between a group of redundant routers.
GLBP for IPv6	This is a Cisco-proprietary FHRP that provides the same functionality of GLBP, but in an IPv6 environment. GLBP for IPv6 provides automatic router backup for IPv6 hosts configured with a single default gateway on a LAN. Multiple first-hop routers on the LAN combine to offer a single virtual first-hop IPv6 router while sharing the IPv6 packet forwarding load.
ICMP Router Discovery Protocol (IRDP)	Specified in RFC 1256, IRDP is a legacy FHRP solution. IRDP allows IPv4 hosts to locate routers that provide IPv4 connectivity to other (nonlocal) IP networks.